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Evaluating Physics-Informed Learning for Fluid Dynamics

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Which is the better way to teach a network fluid flow: simulation data or physics?

A trained neural network predicts a flow in milliseconds instead of running a slow simulation. A data-driven network (data NN) learns from simulation results, so a simulation must run first. A physics-informed network (PINN) [3] learns only from the flow equations: no data, but more computing. A hybrid learns from both.

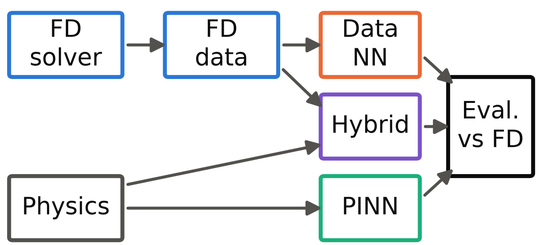
RQ1 Which is more accurate?

RQ2 Which obeys the physics better?

RQ3 Which is more reliable across seeds and flow speeds?

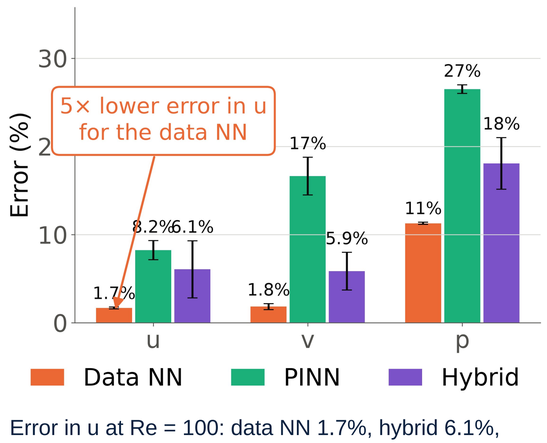
RQ4 What does each cost?

All three networks were built and scored the same way; only what they learn from differs.



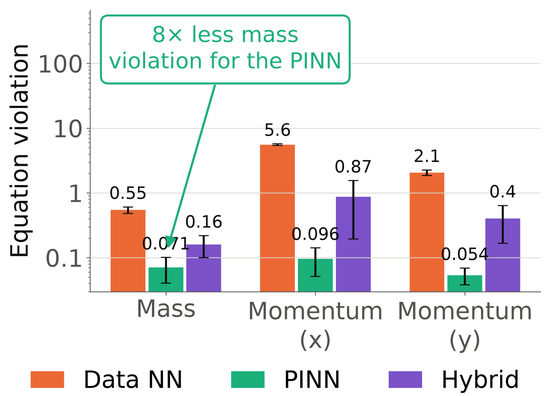
Test flow: fluid in a box driven by a sliding lid. Reference: a finite-difference (FD) simulation that matches published results [1, 2]. All three: 17,091 parameters, 20,000 training steps, 3-5 seeds, scored on points and times never seen in training.

RQ1: The data NN matches the simulation 5× more closely.



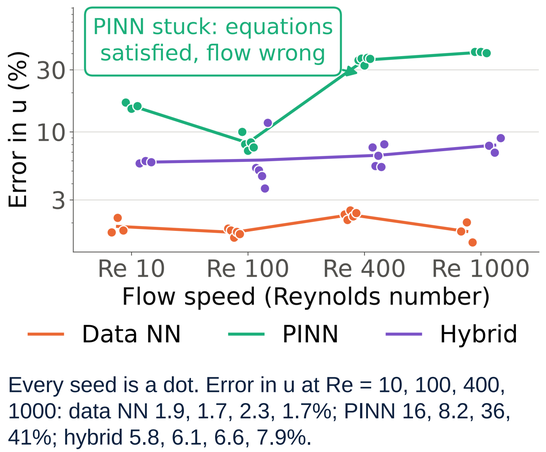
Error in u at Re = 100: data NN 1.7%, hybrid 6.1%, PINN 8.2% (mean of 5 seeds each). One hybrid seed ends on a spike (4.1% → 12% in its last 500 steps); like every run, it is scored at its final step.

RQ2: The PINN obeys the flow equations 8× more closely.



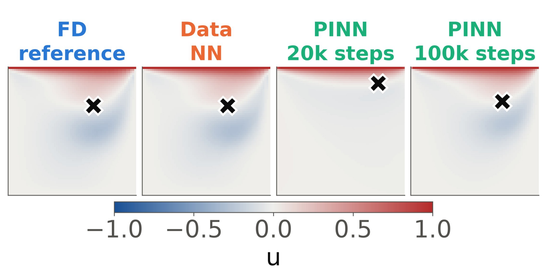
Violation of mass conservation: PINN 0.071, hybrid 0.16, data NN 0.55. Without its physics loss the hybrid has 0.45 (error in u 3.0% vs 5.0%, same seeds).

RQ3: As the flow speeds up, only the PINN breaks down.



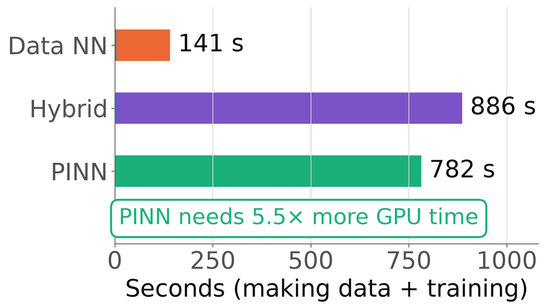
Every seed is a dot. Error in u at Re = 10, 100, 400, 1000: data NN 1.9, 1.7, 2.3, 1.7%; PINN 16, 8.2, 36, 41%; hybrid 5.8, 6.1, 6.6, 7.9%.

RQ3: At Re = 400 the PINN obeys the equations but needs more training, or a head start, to get the flow right.



Flow at t = 5 (× = vortex). The PINN conserves mass better than the data NN (0.03 vs 0.55) yet puts the vortex by the lid. Error in u: 35% → 14% with 5× more steps (3 seeds); 36% → 15% starting from the trained Re = 100 PINN (5 seeds; a random start is at 21% after the same 40,000 steps). The hybrid gets the flow right (6.6%) and conserves mass far better than the data NN (0.07); trained 5× longer it reaches 4.4% (1 seed).

RQ4: The PINN needs no data, but 5.5× more training time.



The simulation took only 3.9 s, so data was cheap; the hybrid pays for both. GPU memory: 708 MB (PINN) vs 122 MB (data NN).

No method wins on everything; each trades accuracy, physics and cost.

- Data NN: most accurate and cheapest, when simulation data already exists.
- PINN: needs no data and obeys the physics best, but costs more and fails as the flow speeds up.
- Hybrid: accurate at every flow speed (7.9% vs 41% for the PINN at Re = 1000) and far closer to the physics than the data NN, but needs data and the most training.
- Reusing a trained network helps: a PINN started from the Re = 100 one more than halved its error at Re = 400.

[1] Marchi, C. H., Suero, R., and Araki, L. K. 2009. The lid-driven square cavity flow: numerical solution with a 1024 x 1024 grid. J. Braz. Soc. Mech. Sci. Eng. 31, 3, 186–198.

[2] Hou, S., Zou, Q., Chen, S., Doolen, G., and Cogley, A. C. 1995. Simulation of cavity flow by the lattice Boltzmann method. J. Comput. Phys. 118, 2, 329–347.

[3] Raissi, M., Perdikaris, P., and Karniadakis, G. E. 2019. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707.

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